

PADHAISHURU

GATE Preparation Platform

GATE EC

Electronics & Communication Engineering

TREND-BASED MOCK PAPER

— 04 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	9 subjects

Difficulty Distribution

Easy: 19 questions (29%)

Moderate: 34 questions (52%)

Difficult: 12 questions (18%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE EC — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

2M [-0.66]

[General Aptitude]

Five years ago, the ratio of Aman's age to his father's age was 1:4, and five years from now, the ratio will be 2:5. What was his father's age when Aman was born?

- (A) 28 years
- (B) 30 years
- (C) 35 years
- (D) 32 years

Q2.

3M [-0.66]

[General Aptitude]

Which one of the following plots represents $() = - ||$, where $||$ is a non-zero real number? Note: The figures shown are representative.

- (A) A
- (B) B
- (C) C
- (D) D

Q3.

2M [-0.66]

[General Aptitude]

Sum of first 100 natural numbers:

- (A) 1000
- (B) 5050
- (C) 4950
- (D) 5000

Q4.

2M [-0.66]

[General Aptitude]

Choose opposite of 'benevolent':

- (A) Hostile
- (B) Kind
- (C) Generous
- (D) Helpful

Q5.

2M [-0.66]

[General Aptitude]

For a real number $x > 1$, $\frac{1}{x} \log 2 + \log 3 + \log 4 = 1$ The value of x is

- (A) 4
- (B) 12
- (C) 24
- (D) 36

Q6.

2M [no -ve]

[General Aptitude]

P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms. All of them agree that the siblings should not share the same room. If S and Q are single children, and the room allocations shown below are acceptable to all, PR TS Q PQ RT S then, which one of the given options is the siblings?

Enter your answer here

Q7.

3M [-0.66]

[General Aptitude]

The greatest prime factor of (3199 - 3196) is

- (A) 13
- (B) 17
- (C) 3
- (D) 11

Q8.

2M [-0.66]

[General Aptitude]

If ` denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to [bored _____ weary]. Which one of the given options is appropriate to fill the blank?

- (A) jaded
- (B) baffled
- (C) dead
- (D) worsted

Q9.

2M [no -ve]

[General Aptitude]

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after `Iteration ' is: Note: The figures shown are representative. 1 Iteration 0: Iteration 1: 1
1 Length of each segment: 3 3 1 1 Iteration 2: 9 Length of each segment: 9

- (A) A
- (B) B
- (C) C
- (D) D

Q10.

2M [no -ve]

[General Aptitude]

The 12 musical notes are given as , #, , #, , , #, , #, , #, and . Frequency of each note is 122 times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes # and is:

- (A) 62
- (B) 2
- (C) 42
- (D) 2

Q11.

2M [-0.66]

[General Aptitude]

Choose synonym of 'trivial':

- (A) Significant
- (B) Insignificant
- (C) Important
- (D) Major

Q12.

2M [-0.66]

[General Aptitude]

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above sentence.

- (A) will be
- (B) can be
- (C) am going to be
- (D) could have been

Q13.

2M [no -ve]

[Engineering Mathematics]

Integral of $1/x$ dx:

Enter your answer here

Q14.

1M [no -ve]

[Engineering Mathematics]

$i^4 =$:

Enter your answer here

Q15.

1M [no -ve]

[Engineering Mathematics]

$P(A \cap B) = ?$

Enter your answer here

Q16.

2M [-0.66]

[Network, Signals & Systems]

In the given circuit, the current I_x (in mA) is _____. 1000 Ω $\rightarrow$ 5 mA 1k 1k 1k Ω I_x 2 mA

- (A) A
- (B) B
- (C) C
- (D) D

Q17.

2M [no -ve]

[Network, Signals & Systems]

Two resistors are connected in a circuit loop of area 5 m², as shown in the figure below. The circuit loop is placed on the x-y plane. When a time-varying magnetic flux, with flux-density (B) = 0.5 (in Tesla), is applied along the positive z-axis, the magnitude of current I (in Amperes, rounded off to two decimal places) in the loop is _____.

Enter your answer here

Q18.

1M [no -ve]

[Network, Signals & Systems]

Z-parameter conditions:

Enter your answer here

Q19.

2M [-0.66]

[Network, Signals & Systems]

Let $\mathbb{R}$ and $\mathbb{C}$ denote the set of real numbers and the three dimensional vector space over it, respectively. The value of α for which the set of vectors $\{[2 \ -3], [3 \ -1 \ 3], [1 \ -5 \ 7]\}$ does not form a basis of $\mathbb{C}$ is _____.

Electronics and Communication Engineering (EC)

- (A) A
- (B) B
- (C) C
- (D) D

Q20.

2M [-0.66]

[Network, Signals & Systems]

A 3-input majority logic gate has inputs A , B , and C . The output of the gate is logic '1' if two or more of the inputs are logic '1'. The output is logic '0' if two or more of the inputs are logic '0'. Which one of the following options is a Boolean expression of the output ?

- (A) A
- (B) B
- (C) C
- (D) D

Q21.

1M [-0.33]

[Network, Signals & Systems]

h-parameter represents:

- (A) Open circuit
- (B) Short circuit
- (C) Hybrid
- (D) Inverse

Q22.

2M [no -ve]

[Network, Signals & Systems]

An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm. Copper has a conductivity of 5.8×10^7 S/m. The current (in Amperes, rounded off to two decimal places) flowing through the wire is _____.

Enter your answer here

Q23.

1M [-0.33]

[Electronic Devices]

n-type majority carriers:

- (A) Holes
- (B) Electrons
- (C) Both equal
- (D) None

Q24.

1M [-0.33]

[Electronic Devices]

MOSFET is:

- (A) BJT
- (B) Unipolar
- (C) Bipolar
- (D) None

Q25.

1M [-0.33]

[Electronic Devices]

SCR has:

- (A) 2 terminals
- (B) 3
- (C) 4
- (D) 5

Q26.

1M [-0.33]

[Electronic Devices]

Zener diode used for:

- (A) Rectification
- (B) Voltage regulation
- (C) Amplification
- (D) None

Q27.

1M [-0.33]

[Analog Circuits]

Common emitter has:

- (A) High input Z
- (B) Low input Z
- (C) Zero gain
- (D) Unity

Q28.

1M [no -ve]

[Analog Circuits]

;" —â \$¥B —3

Enter your answer here

Q29.

1M [-0.33]

[Analog Circuits]

Ideal op-amp open-loop gain:

- (A) 0
- (B) 1
- (C) Infinite
- (D) -1

Q30.

1M [no -ve]

[Analog Circuits]

FET is:

Enter your answer here

Q31.

1M [-0.33]

[Analog Circuits]

RC oscillator frequency:

- (A) $f=1/(2\pi RC)$
- (B) $f=1/RC$
- (C) $f=1/(4\pi RC)$
- (D) $f=2\pi RC$

Q32.

1M [-0.33]

[Digital Circuits]

4-bit binary counter counts:

- (A) 0-15
- (B) 0-9
- (C) 1-16
- (D) 1-10

Q33.

1M [-0.33]

[Digital Circuits]

D flip-flop follows input after:

- (A) Same instant
- (B) One clock
- (C) Two clocks
- (D) Never

Q34.

1M [-0.33]

[Digital Circuits]

ROM is:

- (A) Read only
- (B) Read write
- (C) Volatile
- (D) Cache

Q35.

1M [-0.33]

[Digital Circuits]

TTL stands for:

- (A) Transistor-Transistor Logic
- (B) Tunnel Transistor Logic
- (C) Totem Pole Logic
- (D) None

Q36.

1M [no -ve]

[Control Systems]

Routh-Hurwitz determines:

*Enter your answer here***Q37.**

1M [-0.33]

[Control Systems]

Lead compensator:

- (A) Improves steady state
- (B) Improves transient
- (C) Both
- (D) None

Q38.

1M [-0.33]

[Control Systems]

Nyquist criterion uses:

- (A) Polar plot
- (B) Root locus
- (C) Routh
- (D) Bode

Q39.

1M [no -ve]

[Control Systems]

Transfer function is defined:

*Enter your answer here***Q40.**

1M [-0.33]

[Communication Systems]

AM stands for:

- (A) Amplitude Modulation
- (B) Angle Modulation
- (C) Analog Modulation
- (D) None

Q41.

1M [no -ve]

[Communication Systems]

PCM stands for:

*Enter your answer here***Q42.**

1M [-0.33]

[Communication Systems]

FM has:

- (A) Constant amplitude
- (B) Variable
- (C) Both
- (D) None

Q43.

1M [-0.33]

[Communication Systems]

ASK modulation varies:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q44.

1M [-0.33]

[Electromagnetics]

Maxwell has how many equations:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q45.

1M [-0.33]

*[Electromagnetics]*Characteristic impedance $Z_0 =$:

- (A) $\sqrt{L/C}$
- (B) L/C
- (C) C/L
- (D) $\sqrt{LC}$

Q46.

1M [-0.33]

[Electromagnetics]

Waveguide supports:

- (A) TEM
- (B) TE/TM
- (C) Both
- (D) None

Q47.

1M [-0.33]

[Electromagnetics]

Skin depth increases with:

- (A) Frequency
- (B) Conductivity
- (C) Both
- (D) Neither

Q48.

2M [-0.66]

[Network, Signals & Systems]

In the circuit below, the opamp is ideal. $R_1 = C R R + R C$ If the circuit is to show sustained oscillations, the respective values of 1 and the corresponding frequency of oscillation are _____.

- (A) 29 and $1/(26)$
- (B) 2 and $1/(2)$
- (C) 29 and $1/(2)$
- (D) 2 and $1/(26)$

Q49.

2M [-0.66]

[Network, Signals & Systems]

Which of the following statements is/are TRUE with respect to ideal MOSFET- based DC-coupled single-stage amplifiers having finite load resistors?

- (A) The common-gate amplifier has an infinite input resistance
- (B) The common-source amplifier has an infinite input resistance
- (C) The input and output voltages of the common-source amplifier are in phase
- (D) The input and output voltages of the common-drain amplifier are in phase

Q50.

2M [-0.66]

[Network, Signals & Systems]

For a causal discrete-time LTI system with transfer function $(z) = (z^2 + 3z + 13) / (z^2 - 13z + 13)$ - which of the following statements is/are true?

- (A) The system is stable.
- (B) The system is a minimum phase system.
- (C) The initial value of the impulse response is 2.
- (D) The final value of the impulse response is 0.

Q51.

1M [-0.33]

[Network, Signals & Systems]

FS exists for:

- (A) Periodic
- (B) Aperiodic
- (C) Both
- (D) None

Q52.

2M [no -ve]

[Network, Signals & Systems]

The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the points A, B, and C mark three different positions. Which of the following statements is/are true?

Enter your answer here

Q53.

2M [-0.66]

[Network, Signals & Systems]

Consider a system S represented in state space as $\dot{x} = 0 - 2x + 1r$, $y = 2 - 5x$. Which of the state space representations given below has/have the same transfer function as that of S?

- (A) $\dot{x} = 0 - 10$, $y = 1 - 2x$
- (B) $\dot{x} = 0 - 1x + 1r$, $y = 0 - 2x$
- (C) $\dot{x} = -1 - 0$, $y = 1 - 1x$
- (D) $\dot{x} = -1 - 0x + 1r$, $y = 1 - 2x$

Q54.

2M [-0.66]

[Network, Signals & Systems]

Two fair dice (with faces labeled 1, 2, 3, 4, 5, and 6) are rolled. Let the random variable denote the sum of the outcomes obtained. The expectation of is _____ (rounded off to two decimal places).

- (A) A
- (B) B
- (C) C
- (D) D

Q55.

1M [-0.33]

[Electronic Devices]

n-type majority carriers:

- (A) Holes
- (B) Electrons
- (C) Both equal
- (D) None

Q56.

2M [-0.66]

[Network, Signals & Systems]

The general form of the complementary function of a differential equation is given by $y(t) = (At + B)e^{-2t}$, where A and B are real constants determined by the initial condition. The corresponding differential equation is _____.

- (A) $\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 4y = f(t)$
- (B) $\frac{d^2 y}{dt^2}$
- (C) $\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2y = f(t)$
- (D) $\frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y = f(t)$

Q57.

2M [-0.66]

[Network, Signals & Systems]

An amplitude modulator has output (in Volts) $s(t) = A \cos(400t) + B \cos(360t) + B \cos(440t)$. The carrier power normalized to 1 resistance is 50 Watts. The ratio of the total sideband power to the total power is $1/9$. The value of (in Volts, rounded off to two decimal places) is _____.

- (A) A
- (B) B
- (C) C
- (D) D

Q58.

2M [no -ve]

[Network, Signals & Systems]

In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time constant (in seconds) of the circuit for $t > 0$ is _____. $t=0$ 3H S 5A 2 4 4

Enter your answer here

Q59.

2M [no -ve]

[Network, Signals & Systems]

For a causal discrete-time LTI system with transfer function $H(z) = (2z^2 + 3z + 13) / (z^2 - 13z + 13)$ - which of the following statements is/are true?

Enter your answer here

Q60.

2M [-0.66]

[Network, Signals & Systems]

Let $X(t) = A \cos(2\pi f_0 t + \phi)$ be a random process, where amplitude A and phase ϕ are independent of each other, and are uniformly distributed in the intervals $[-2, 2]$ and $[0, 2\pi]$, respectively. $X(t)$ is fed to an 8-bit uniform mid-rise type quantizer. Given that the autocorrelation of $X(t)$ is $R_X(\tau) = 2 \cos(2\pi f_0 \tau)$, the signal to 3 quantization noise ratio (in dB, rounded off to two decimal places) at the output of the quantizer is _____.

- (A) A
- (B) B
- (C) C
- (D) D

Q61.

2M [-0.66]

[Network, Signals & Systems]

As shown in the circuit, the initial voltage across the capacitor is 10 V, with the switch being open. The switch is then closed at $t = 0$. The total energy dissipated in the ideal Zener diode ($V_Z = 5$ V) after the switch is closed (in mJ, rounded off to three decimal places) is _____. 0 10 k 10 F $+ = 5$ V $(0^-) = 10$ V -

Electronics and Communication Engineering (EC) Q.36 to Q.65 Carry TWO marks Each

- (A) A
- (B) B
- (C) C
- (D) D

Q62.

2M [-0.66]

[Network, Signals & Systems]

The opamps in the circuit shown are ideal, but have saturation voltages of ± 10 V. Assume that the initial inductor current is 0 A. The input voltage (V_i) is a triangular signal with peak voltages of ± 2 V and time period of 8 μ s. Which one of the following statements is true?

- (A) V_{O1} is delayed by 2 μ s relative to V_i , and V_{O2} is a triangular waveform.
- (B) V_{O1} is not delayed relative to V_i , and V_{O2} is a trapezoidal waveform.
- (C) V_{O1} is not delayed relative to V_i , and V_{O2} is a triangular waveform.
- (D) V_{O1} is delayed by 1 μ s relative to V_i , and V_{O2} is a trapezoidal waveform.

Q63.

2M [no -ve]

[Network, Signals & Systems]

and are Bernoulli random variables taking values in $\{0,1\}$. The joint probability mass function of the random variables is given by: $P(X=0, Y=0) = 0.06$ $P(X=0, Y=1) = 0.14$ $P(X=1, Y=0) = 0.24$ $P(X=1, Y=1) = 0.56$ The mutual information ($I(X; Y)$) is _____ (rounded off to two decimal places). Electronics and Communication Engineering (EC)

Enter your answer here

Q64.

1M [-0.33]

[Control Systems]

Lead compensator:

- (A) Improves steady state
- (B) Improves transient
- (C) Both
- (D) None

Q65.

2M [no -ve]

[Network, Signals & Systems]

Consider a non-negative function $f(x)$ which is continuous and bounded over the interval $[2, 8]$. Let M and m denote, respectively, the maximum and the minimum values of $f(x)$ over the interval. Among the combinations of M and m given below, choose the one(s) for which the inequality $28(M - m)$ is guaranteed to hold.

- (A) $M = 5, m = 7$
- (B) $M = 6, m = 5$
- (C) $M = 7, m = 6$
- (D) $M = 7, m = 5$